



FREDYS MATOS BORGES

HARDWARE ENGINEER



Professional Profile

Hardware and PCB Design Engineer with extensive experience combining electronic design, embedded systems, and validation in real-world operating environments. Specialized in PCB design for low-power systems, automation, and electronics.



Work Experiences

PCB / HARDWARE ENGINEER

(2025-2026)

InGnia Technology, Havana, Cuba (Freelance)

- Design and development of PCB schematics and layouts for embedded control systems, integrating microcontrollers, sensors, power management circuits, and communication modules (Wi-Fi, Bluetooth, Zigbee).
- Participation in complete hardware development cycles, from concept and schematic design to prototyping, testing, and DFM reviews.
- Creation and maintenance of design documentation, including bills of materials (BOM), assembly drawings, and test protocols.
- Collaboration with firmware and product teams to ensure seamless system integration, as well as functionality and performance in real-world environments.

HARDWARE ENGINEER

(2025)

CIRCE Smart Navigation Solutions, Barcelona, Spain (Freelance)

- Development of a compact wearable system aimed at tracking and monitoring applications.
- Schematic design and multilayer PCB layout in Altium Designer.
- Integration of GNSS module for positioning.
- Implementation of environmental and motion sensors (IMU and temperature).
- Power architecture design for battery operation and energy consumption optimization.
- Integration of wireless communications (Wi-Fi / Bluetooth and Sub-GHz radio).
- Design for hardware debugging, programming, and validation.

HARDWARE / AUTOMATION ENGINEER

(2022-2025)

Soluciones Electro-Automáticas SRL, Havana, Cuba (Hybrid)

- Hardware and PCB design for residential automation systems.
- PCB design optimization reducing manufacturing costs by 15%.
- Improved system reliability through application of DFM best practices.
- Participation in the full cycle: design → prototype → documentation → technical support.

PROJECTS ENGINEER

(2019-2022)

Almacenes Universales S.A., Havana, Cuba (Hybrid)

- Technical design of over 50 electronic security and home automation projects.
- Preparation of schematics, technical drawings, and selection of electronic components.
- Design optimization that reduced installation times by 20%.
- Definition of component selection processes that improved system compatibility by 25%.

ROBOTICS RESEARCH ANALYST

(2019-2022)

Technological University of Havana, Havana, Cuba (Hybrid)

- Proposal of a mobile robot prototype for automating logistics processes in warehouses using automated picking systems and e-commerce.
- Analysis of existing commercial solutions, identifying efficiency improvements of up to 35%.
- Technical documentation of applied research for future developments.



References

SADIEL DE LA FE

Ph.D. in Telecommunications and Electronics Engineering

CEO & Co-founder at Circe Smart Navigations

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NASIEL GARCÍA FERNÁNDEZ

Telecommunications and Electronics Engineer

Senior Observability Engineer en Veriff

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Valid residence and
work permit for Spain



Education

Technological University
of Havana

Automation Engineer (2012-2017)



Skills

HARDWARE & PCB

- Multilayer PCB Design
- DFM / DFT
- Low-Power Design
- Prototyping and Validation

TOOLS

- KiCad
- Altium Designer
- Proteus
- EasyEDA
- LTspice
- Oscilloscope
- Multimeter

EMBEDDED

- ESP32
- STM32
- PICs
- C/C++
- MicroPython



Languages

- Spanish (Native)
- English (B2)